Advanced Linear Algebra: Foundations to Frontiers

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8.2.3 Relation to Splitting Methods

8.2.3 Gauss Seidel as a descent method, Part 1

fit width

Let us pick some really simple search directions in the right-most algorithm in Homework 8.2.2.2: $p^{(k)} = e_{k \bmod n}$, which cycles through the standard basis vectors.

Homework 8.2.3.1. For the right-most algorithm in Homework 8.2.2.2, show that if $p^{(0)} = e_0$, then

$$\chi_0^{(1)} = \chi_0^{(0)} + \frac{1}{\alpha_{0,0}} \left(\beta_0 - \sum_{j=0}^{n-1} \alpha_{0,j} \chi_j^{(0)} \right) = \frac{1}{\alpha_{0,0}} \left(\beta_0 - \sum_{j=1}^{n-1} \alpha_{0,j} \chi_j^{(0)} \right).$$

▾ Solution

- $p^{(0)} = e_0$.
- $p^{(0)\,T} A p^{(0)} = e_0^T A e_0 = \alpha_{0,0}$ (the $(0,0)$ element in A, not to be mistaken for α_0).
- $r^{(0)} = Ax^{(0)} - b$.
- $p^{(0)\,T} r^{(0)} = e_0^T (b - Ax^{(0)}) = e_0^T b - e_0^T A x^{(0)} = \beta_0 - \tilde a_0^T x^{(0)}$, where $\tilde a_k^T$ denotes the kth row of A.
- $x^{(1)} = x^{(0)} + \alpha_0 p^{(0)} = x^{(0)} + \frac{p^{(0)\,T} r^{(0)}}{p^{(0)\,T} A p^{(0)}} e_0 = x^{(0)} + \frac{\beta_0 - \tilde a_0^T x^{(0)}}{\alpha_{0,0}} e_0$. This means that only the first element of $x^{(0)}$ changes, and it changes to

$$\chi_0^{(1)} = \chi_0^{(0)} + \frac{1}{\alpha_{0,0}} \left(\beta_0 - \sum_{j=0}^{n-1} \alpha_{0,j} \chi_j^{(0)} \right) = \frac{1}{\alpha_{0,0}} \left(\beta_0 - \sum_{j=1}^{n-1} \alpha_{0,j} \chi_j^{(0)} \right).$$

This looks familiar...

8.2.3 Gauss Seidel as a descent method, Part 2

fit width

Careful contemplation of the last homework reveals that this is exactly how the first element in vector x, χ_0, is changed in the Gauss-Seidel method!

Ponder This 8.2.3.2. Continue the above argument to show that this choice of descent directions yields the Gauss-Seidel iteration.